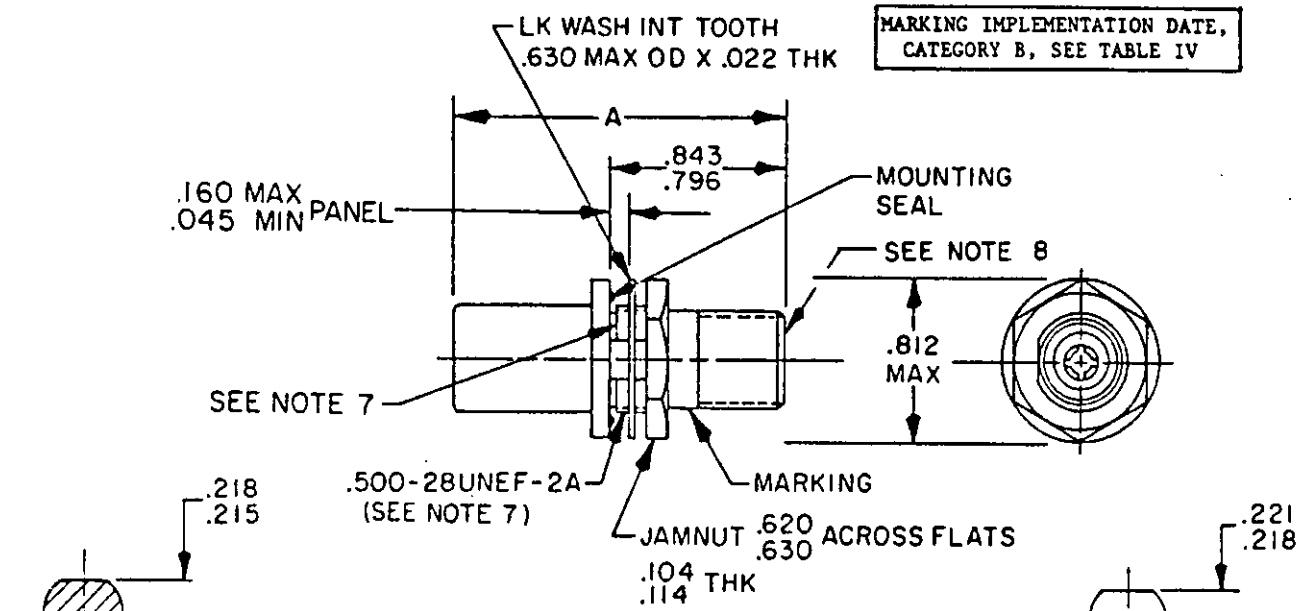
MILITARY SPECIFICATION SHEET

# CONNECTORS, RECEPTACLE, ELECTRICAL, COAXIAL, RADIO FREQUENCY

## (SERIES TNC (CABLED), SOCKET CONTACT, JAM NUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .022 | 0.56 | .495 | 12.57 |
| .045 | 1.14 | .500 | 12.70 |
| .104 | 2.64 | .505 | 12.83 |
| .114 | 2.90 | .510 | 12.95 |
| .160 | 4.06 | .620 | 15.75 |
| .215 | 5.46 | .630 | 16.00 |
| .218 | 5.54 | .796 | 20.22 |
| .221 | 5.61 | .812 | 20.62 |
| .493 | 12.52 | .843 | 21.41 |

NOTES:
1. Dimensions are in inches.
2. For dimension A, see table I and III.
3. Metric equivalents are given for general information only.
4. Wrench flats are to accommodate standard wrench opening in accordance with FED-STD-H28, appendix 10.
5. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
6. All undimensioned pictorial configurations are for reference purposes only.
7. Full threads to within .063 (1.60 mm) of shoulder.
8. Series TNC, socket contact interface in accordance with MIL-STD-348.

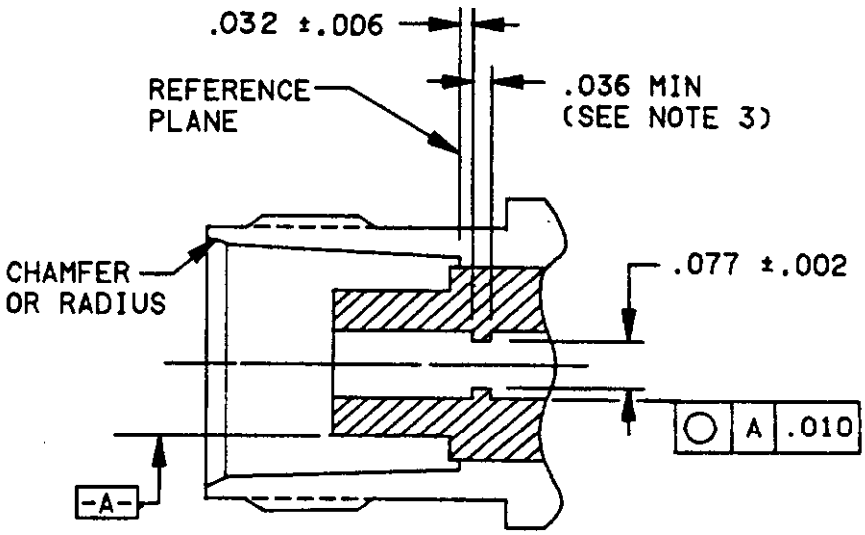
FIGURE 1. General configuration.

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AMSC N/A
FSC 5935

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## CATEGORY D

| Inches | mm |
|---|---|
| .002 | 0.05 |
| .006 | 0.15 |
| .010 | 0.25 |
| .036 | 0.91 |
| .077 | 1.96 |
| .234 | 5.94 |
| .4375 | 11.112 |

NOTES:
1. Metric equivalents are given for general information only.
2. All undimensioned pictorial configurations are for reference purpose only.
3. Chamfer is optional.
4. Dimensions are in inches.

FIGURE 2. Category D captivation detail.

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TABLE I. Dash numbers, cross reference, and dimensions.

| Dash number 1/ | Applicable cable M17/ # | Dimensions | Inches (millimeters) maximum |
|---|---|---|---|
| Category A - Field serviceable (no special tools required) | | | |
| 0101 3/ | 028-RG058<br>084-RG223*<br>128-RG400<br>060-RG142@ | A | 1.250 (31.75) |
| 0102 3/ | 29-RG59*Δ<br>030-RG062Δ<br>90-RG71Δ<br>97-RG210Δ | | |
| 0103 3/ | 054-RG122* | | |
| 0104 3/ | 111-RG303@* | | |
| 0117 3/ | 110-RG302*@Δ | | |
| 0018 3/ | 113-RG316*@<br>119-RG174 | | |
| Category C - Field replaceable (MIL-C-22520 crimp tool) 2/ See footnote symbol next to applicable cable for crimp die | | | |
| 0010 3/ | 028-RG058*~<br>111-RG303@~ | A | 1.625 (41.28) |
| 0011 3/ | 084-RG223*~<br>128-RG400~<br>060-RG142@~ | | |
| 0012 3/ | 29-RG59*Δ><br>030-RG062Δ><br>97-RG210Δ> | | |
| 0013 3/ | 054-RG122*& | | |
| 0014 3/ | 90-RG71*Δ> | | |
| 0021 3/ | 110-RG302*>@Δ | | |
| 0022 3/ | 113-RG316@*Σ<br>119-RG174Σ | | |
| 0023 3/ | 95-RG180*Δ& | | |

See footnotes at end of table.

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TABLE I. Dash numbers, cross reference, and dimensions - Continued.

| Dash number 1/ | Applicable cable M17/ # | Dimensions | Inches (millimeters) maximum |
|---|---|---|---|
| Category D - Field replaceable - Defined piece parts 2/4/5/ | | | |
| 0501 3/ | 054-RG122* | A | 1.468 (37.36) |
| 0502 3/ | 95-RG180*∆ | | |
| 0503 3/ | 060-RG142*<br>128-RG400 | | |
| 0504 3/ | 028-RG058* | | |

1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.

3/ These connectors have captivated center contacts.

4/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

5/ Not to be used in Army equipment.

\# The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and ∆.

@ Cable to be used for the +200°C temperature cycling tests.

∆ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

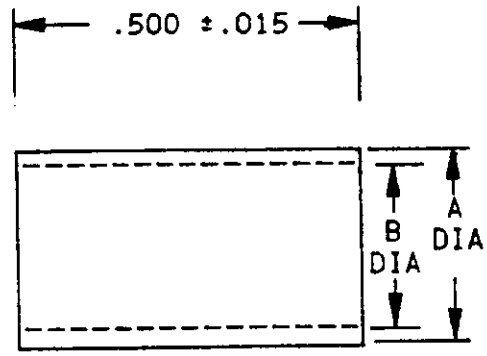
Σ M22520/5-35 Closure B or M22520/5-03 Closure A

& M22520/5-41 Closure B or M22520/5-05 Closure B
M22520/5-09 Closure A

~ M22520/5-19 Closure B or M22520/5-05 Closure A
M22520/5-11 Closure A
M22520/5-57 Closure A

> M22520/5-19 Closure A or M22520/5-07 Closure A
M22520/5-13 Closure A
M22520/5-59 Closure A

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:51Z
Check the source to verify that this is the current version before use.

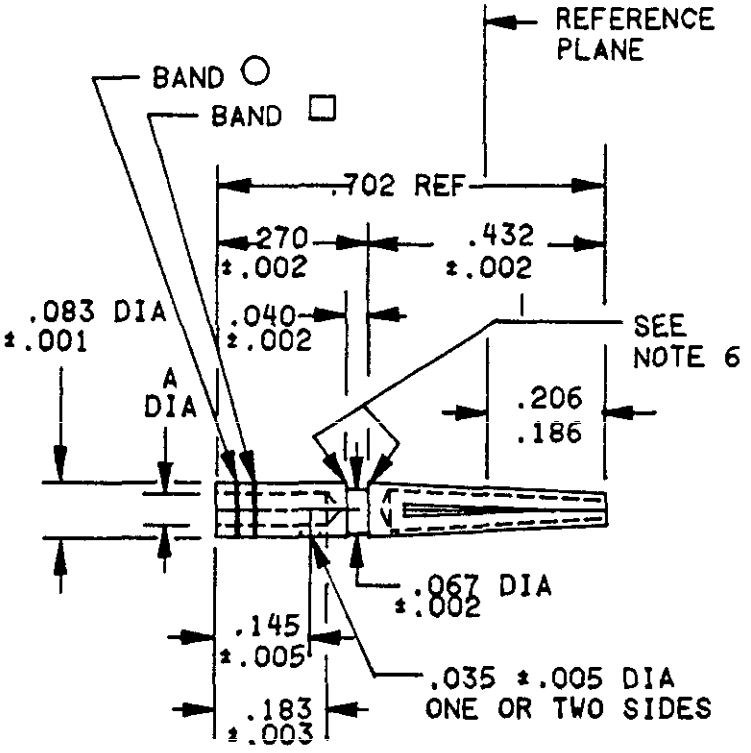



| Dash no. | Ferrule no. 2/ | A +.001 | B ±.001 | Basic crimp tool 1/ | Crimp die or positioner M22520/5- |
|---|---|---|---|---|---|
| 0501<br>0502 | 28-50 | .212 | .175 | M22520/1-01 | 05, 41<br>Closure B<br>or<br>9<br>Closure A |
| 0503 | 28-51 | .250 | .220 | | 05, 11, 57<br>Closure A<br>or<br>19<br>Closure B |
| 0504 | 28-52 | .245 | .206 | | |

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

FIGURE 3. Contact and ferrule dimensions for category D only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:51Z
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| Inches | mm |
|---|---|
| .001 | 0.03 |
| .002 | 0.05 |
| .003 | 0.08 |
| .005 | 0.13 |
| .015 | 0.38 |
| .017 | 0.43 |
| .033 | 0.84 |
| .035 | 0.89 |
| .040 | 1.02 |
| .043 | 1.09 |
| .067 | 1.70 |
| .083 | 2.11 |
| .145 | 3.68 |
| .175 | 4.45 |
| .183 | 4.65 |
| .186 | 4.72 |
| .206 | 5.23 |
| .212 | 5.38 |
| .220 | 5.59 |
| .245 | 6.22 |
| .250 | 6.35 |
| .270 | 6.86 |
| .432 | 10.97 |
| .500 | 12.70 |
| .702 | 17.83 |

## CENTER CONTACT

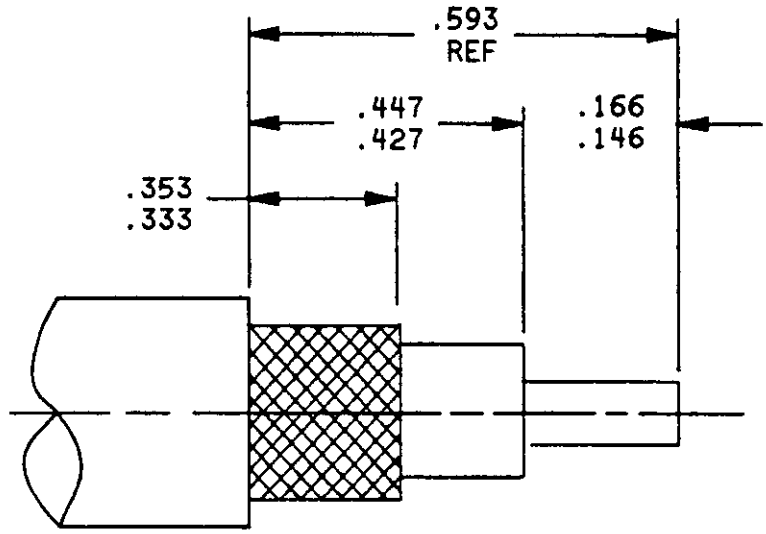
| Dash no. | Contact no. 2/ | A +.001 -.002 | Basic crimp tool 1/ | Crimp die or positioner | Crimp tensile min | Color band | Color band |
|---|---|---|---|---|---|---|---|
| 0501 | 28-12 | .033 | M22520/1-01 | M22520/1-12 | 10 lbs | Orn | Blue |
| 0502 | 28-11 | .017 | | | 6 lbs | Blue | |
| 0503<br>0504 | 28-10 | .043 | M22520/1-01 | M22520/1-12 | 20 lbs | Red | |

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Contact material shall be phosphor bronze or copper beryllium.
4. Crimp tensile test shall be in accordance with MIL-C-39029.
5. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
6. .003 (0.08 mm) maximum break.

FIGURE 3. Contact and ferrule dimensions for category D only - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:51Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .146 | 3.71 |
| .166 | 4.22 |
| .333 | 8.46 |
| .353 | 8.97 |
| .427 | 10.85 |
| .447 | 11.35 |
| .593 | 15.06 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:51Z
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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

500 volts rms, maximum working voltage at sea level.

125 volts rms, maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque: 4 to 6 inch-pounds.

Mating characteristics:

Reference MIL-STD-348 and figure 2 for dimensions.

Center contact (socket):

Oversize test pin: .057 diameter, minimum (non-closed entry contacts only).

Insertion depth: .125, minimum.

Number of insertions: One.

Insertion force test: Steel test pin diameter .054, minimum.

Test pin finish: 16 microinches.

Insertion force: 2 pounds, maximum.

Withdrawal force test: Steel test pin diameter .052, maximum.

Withdrawal force: 2 ounces, minimum.

Test pin finish: 16 microinches.

Hermetic seal: Not applicable.

Leakage: Connector shall be mounted in mounting hole specified on figure 1 with mating end capped. Test applicable to mounting seal only.

Air pressure: 30 $lbf/in^2$.

Duration: 30 seconds, minimum.

Insulation resistance: Method 302 of MIL-STD-202, test condition B, 5,000 megohms, minimum.

Center contact retention: 6 pounds, minimum axial force. Applicable to captivated-center-contact connectors only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:51Z
Check the source to verify that this is the current version before use.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30, maximum.

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.02+.003 F (F in GHz).

Item 16: VSWR shall be less than 1.02+.003 F (F in GHz).

Second step of VSWR checkout procedure: VSWR shall be less than 1.06+.007 F (F in GHz).

Group B inspection: VSWR shall be less than 1.05+.005 F (F in GHz).

Qualification and group C inspection - VSWR shall not exceed 1.10.

Connector durability: 500 cycles, minimum at 12 cycles per minute, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum:

| | Initial | After environment |
|---|---|---|
| Center contact | 1.5 | 2.0 |
| Outer contact | .2 | Not applicable |
| Braid to body | .1 | Not applicable |

Dielectric withstanding voltage: Method 301 of MIL-STD-202, 1,500 volts rms, minimum at sea level.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see tables I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 375 volts rms, minimum.

Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,000 volts rms tested at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:51Z
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Cable retention force:

Noncrimp assemblies: 40 pounds, minimum.

Crimp assemblies:

10 pounds, minimum for cables .155 - .189 OD.

20 pounds, minimum for cables .190 - .229 OD.

30 pounds, minimum for cables .230 - .249 OD.

40 pounds, minimum for cables .250 OD and larger.

Coupling mechanism retention force: Not applicable.

RF leakage: -60 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss:

.18 dB maximum tested at 9 GHz.

.06 $\sqrt{F\ (GHz)}$ dB maximum tested at 3 GHz and 6 GHz.

Part number: M39012/28 (dash number from table I or "B" number from table III).

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TABLE II. Group qualification.

| Group | Submission and qualification of any of the following connectors | Qualifies the following connectors |
|---|---|---|
| | M39012/ | M39012/ |
| I | 27-0101<br>27-0103<br>27-0104<br>27-0018<br>28-0101<br>28-0103<br>28-0104<br>28-0018<br>29-0101<br>29-0103<br>29-0104<br>29-0018 | 27-0101<br>27-0002<br>27-0103<br>27-0104<br>27-0117<br>27-0018<br>28-0101<br>28-0102<br>28-0103<br>28-0104<br>28-0117<br>28-0018<br>29-0101<br>29-0102<br>29-0103<br>29-0104<br>29-0117<br>29-0018 |
| II | 27-0002<br>27-0117<br>28-0102<br>28-0117<br>29-0102<br>29-0117 | 27-0002<br>27-0117<br>27-0102<br>28-0117<br>29-0102<br>29-0118 |
| III | 27B0005<br>27B0006<br>27B0008<br>27B0015<br>27B0016<br>27B0020<br>28B0005<br>28B0006<br>28B0008<br>28B0015<br>28B0016<br>28B0020<br>29B0005<br>29B0006<br>29B0008<br>29B0015<br>29B0016<br>29B0020 | 27B0005<br>27B0006<br>27B0007<br>27B0008<br>27B0009<br>27B0015<br>27B0016<br>27B0019<br>27B0020<br>28B0005<br>28B0006<br>28B0007<br>28B0008<br>28B0009<br>28B0015<br>28B0016<br>28B0019<br>28B0020<br>29B0005<br>29B0006<br>29B0007<br>29B0008<br>29B0009<br>29B0015<br>29B0016<br>29B0019<br>29B0020 |

See footnotes at end of table.

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TABLE II. Group qualification - Continued.

| Group | Submission and qualification of any of the following connectors M39012/ | Qualifies the following connectors M39012/ |
|---|---|---|
| IV | 27B0007<br>27B0009<br>27B0019<br>28B0007<br>28B0009<br>28B0019<br>29B0007<br>29B0009<br>29B0019 | 27B0007<br>27B0009<br>27B0019<br>28B0007<br>28B0009<br>28B0019<br>29B0007<br>29B0009<br>29B0019 |
| V | 27-0010<br>27-0011<br>27-0013<br>27-0022<br>27-0023<br>28-0010<br>28-0011<br>28-0013<br>28-0022<br>28-0023<br>29-0011<br>29-0013<br>29-0022 | 27-0010<br>27-0011<br>27-0012<br>27-0013<br>27-0014<br>27-0021<br>27-0022<br>27-0023<br>28-0010<br>28-0011<br>28-0012<br>28-0013<br>28-0014<br>28-0021<br>28-0022<br>28-0023<br>29-0010<br>29-0011<br>29-0012<br>29-0013<br>29-0014<br>29-0021<br>29-0022 |
| VI | 27-0012<br>27-0014<br>27-0021<br>28-0012<br>28-0014<br>28-0021<br>29-0012<br>29-0014<br>29-0021 | 27-0012<br>27-0014<br>27-0021<br>28-0012<br>28-0014<br>28-0021<br>29-0012<br>29-0014<br>29-0021 |
| VII | 28-0501<br>28-0503<br>28-0504 | 28-0501<br>28-0502<br>28-0503<br>28-0504 |
| VIII | 28-0502 | 28-0502 |

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE III. Category B - Nonfield replaceable (special tools may be required).

| Not for Air Force or Navy use. For OEM use only. |
|---|

| Dash number 1/ 2/ 3/ | Applicable cable M17/ # | Dimensions | Inches (millimeters) maximum |
|---|---|---|---|
| M39012/28B0005 | 028-RG058* | A | 1.625 (41.28) |
| M39012/28B0006 | 060-RG142@*<br>128-RG400 | | |
| M39012/28B0007 | 29-RG59*Δ<br>030-RG062Δ<br>97-RG210Δ | | |
| M39012/28B0008 | 054-RG122* | | |
| M39012/28B0009 | 90-RG71*Δ | | |
| M39012/28B0015 | 111-RG303*@ | | |
| M39012/28B0016 | 084-RG223* | | |
| M39012/28B0019 | 110-RG302*Δ | | |
| M39012/28B0020 | 113-RG316@*<br>119-RG174 | | |

1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ For maintenance replacements for category B, see table V.

3/ Inactive for new design.

# The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to be used for the +200°C temperature cycling tests.

Δ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

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TABLE IV. Cross reference of part numbers.

| Preferred part number | Substitute for part number or type designation 1/2/ |
|---|---|
| M39012/28- | M39012/28- |
| 0010 | M23329/4-13, M23329/4-15 |
| 0011 | M23329/4-14, M23329/4-16 |
| 0012 | M23329/4-17 |
| 0023 | M23329/4-18 |
| 0101 | 0001 |
| 0102 | 0002 |
| 0103 | 0003 |
| 0104 | 0004 |
| 0117 | 0017 |
| B0005 3/ | 0005 |
| B0006 3/ | 0006 |
| B0007 3/ | 0007 |
| B0008 3/ | 0008 |
| B0009 3/ | 0009 |
| B0015 3/ | 0015 |
| B0016 3/ | 0016 |
| B0017 3/ | 0019 |
| B0020 3/ | 0020 |

1/ The superseded part number or the type designation is for cross reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/28-XXXX shall be used in all cases for marking and identifying the connector.

2/ The basic type designation includes all letter versions of the specified number, e.g. UG-18/U includes UG-18 A/U, UG-18B/U, etc.

3/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

TABLE V. Maintenance replacements for category B.

| Category B number* Inactive for new design | Category C dash number | Category A dash number | Category D dash number |
|---|---|---|---|
| B0005 | 0010 | 0101 | 0504 |
| B0006 | 0011 | 0101 | 0503 |
| B0007 | 0012 | 0102 | --- |
| B0008 | 0013 | 0103 | 0501 |
| B0009 | 0014 | 0102 | --- |
| B0015 | 0010 | 0104 | --- |
| B0016 | 0011 | 0101 | --- |
| B0019 | 0021 | 0117 | --- |
| B0020 | 0022 | 0018 | --- |

*Category B connectors are for original installation only. They will not be stocked or acquired by the Government.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - CR, EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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